

Dynamic Causal Dimensionality in Nonstationary Complex Systems:

Effective-Information Concentration and Evidence from U.S. Power Grids

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Abstract

Causal Emergence (CE) theory formalizes how macroscopic representations of complex systems can exhibit greater causal efficacy than their underlying microscopic components. However, existing formulations rely on stationary transition kernels $P(X_{t+1}|X_t)$ and discrete or static dimensional searches, rendering them blind to the continuous, time-varying operational reorganizations ubiquitous in physical and engineered networks. Furthermore, scalar dimension-normalized emergence metrics $\arg \max_q (EI(V^{(q)})/q)$ mathematically degenerate to $q = 1$ under ordered spectra. Here, we establish the theory and methodology of **Dynamic Causal Dimensionality (DCD)**, resolving the active number of macroscopic causal degrees of freedom through the continuous **Causal Information Spectrum** $e_{i,t} = \frac{1}{2} \ln(1 + \lambda_{i,t})$. We define the continuous **Dynamic Causal Participation Ratio** (DCD_t^{PR}) and **Dynamic Causal Effective Rank** (DCD_t^{entropy}), proving their strict rotational invariance under orthogonal coordinate changes. Measure-theoretic interventional semantics are formalized through an **observational lifting kernel** $\kappa_W(dx | v) \triangleq P_{\text{obs}}(X_t \in dx | W^\top X_t = v)$, proving that empirical localized regressions directly recover the induced macroscopic channel. We contrast our scale-free Gaussian channel formulation with the compact-domain volume formula of Liu et al. (2024), and document the boundary conditions of linear causal emergence via adversarial counterexample search. Benchmarking on canonical nonstationary DGPs (A–I; $R = 100$) establishes an observed false-positive rate of 0/100 (FPR = 0.000, 95% Clopper-Pearson CI [0.000, 0.036]). Applying the framework to 2021–2022 continental U.S. power grid operations (EIA-930; 367,920 BA-hours across Eastern, Western, and ERCOT interconnections) with direction-aware surrogate inference ($B = 1000$ refitted IAAFT surrogates) confirms highly significant macroscopic causal contraction ($p < 0.001$, FDR significant ratio 100.0% for DCD^{PR} across all three interconnections). Generalized Additive Models with cyclic P-

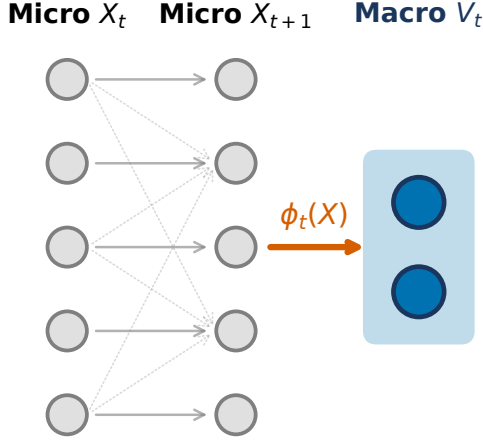
splines and Hansen (1996) supremum-Wald block bootstrap ($B = 2000$) demonstrate smooth non-linear coordination associated with renewable penetration ($R^2 \approx 49\%$), with empirical thresholds at $\hat{\gamma} = 14.5\%$ in ERCOT (sup-Wald $p = 0.585$, 95% CI [13.4%, 40.6%]) and $\hat{\gamma} = 10.7\%$ in Western (sup-Wald $p = 0.658$, 95% CI [10.7%, 37.2%]), indicating continuous adaptation rather than discrete structural breaks. Multi-event analysis reveals acute causal concentration during extreme weather: during Winter Storm Uri, Causal Concentration Gain (CCG_t) surged from 0.086 baseline to 0.244 ($p_{\text{surge}} = 0.0250$), while effective degrees of freedom were resiliently maintained ($\Delta DCD^{\text{PR}} = +0.328$, $p = 0.253$, 0% collapse rate); the multi-event panel test ($T_{\text{Fisher}} = 16.28$, $p = 0.0919$) indicates crisis-specific rather than universal cross-event response. Finally, strictly lookahead-free walk-forward forecasting demonstrates that DCD functions as an interpretable information-theoretic structural coordination diagnostic rather than an unconditional linear forecaster (at $h = 12$ h, an unadjusted deterioration of $\Delta \text{RMSE} = -4.09\%$, $p = 0.0354$ does not survive Benjamini-Hochberg FDR multiplicity correction, $q = 0.1280$).

1 Introduction

Complex dynamical networks—from neural circuits and cellular metabolisms to continental power grids and financial markets—spontaneously organize across disparate spatiotemporal scales [4, 5]. In such systems, macroscopic collective variables often evolve with greater determinism and lower degeneracy than individual microscopic elements [4, 12, 13]. The quantitative theory of **Causal Emergence (CE)**, formalized through interventional information theory [1, 4, 10], provides an exact criterion: a macroscopic representation $V = \phi(X)$ exhibits causal emergence when its Effective Information (EI) exceeds that of the underlying microscopic state X , i.e., $EI(V) > EI(X)$.

Despite fundamental theoretical insights, existing formulations suffer from a critical limitation: *all established*

a | Dynamic Coarse-Graining Mapping



b | Dynamic Causal Emergence (DCE_t)

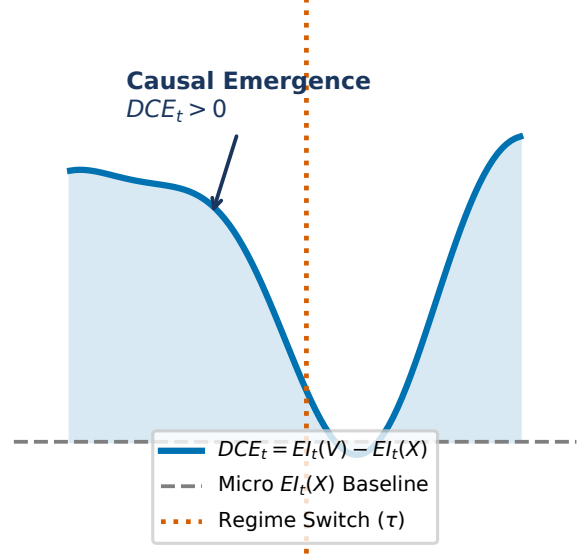


Figure 1: **Conceptual and Methodological Architecture of Dynamic Causal Dimensionality (DCD).** **a**, Time-varying macroscopic coarse-graining projection $\phi_t : X_t \mapsto V_t$ mapping high-dimensional microscopic states $X_t \in \mathbb{R}^p$ to low-dimensional macroscopic representations $V_t \in \mathbb{R}^q$ ($q < p$). **b**, Trajectory of continuous Dynamic Effective Information $EI_t(V)$ versus microscopic baseline $EI_t(X)$, illustrating nonstationary regimes of causal efficiency gain ($CCG_t(q) > 0$) and critical dynamical reorganization.

CE estimators implicitly presume a stationary transition mechanism $P(X_{t+1} | X_t)$. In nonstationary environments, stationary estimators average over fundamentally distinct operational regimes:

- **Renewable Energy Ingestion:** Power grids undergo rapid, weather-driven swings in synchronous inertia and physical topology as inverter-based wind and solar resources displace thermal generators [8, 15].
- **Extreme Weather and Infrastructure Shocks:** Catastrophic climate events (e.g., severe winter freezes, heatwaves) induce structural regime shifts, cascading line trips, and emergency load-shedding [9, 16].
- **Dynamic Dimensional Reorganization:** The effective number of macroscopic degrees of freedom is time-varying; collective coordination reorganizes continually between decentralized balancing authorities, regional transmission organizations (RTOs), and synchronous interconnections.

To address these challenges, this paper develops **Dynamic Causal Dimensionality (DCD)**, contributing:

1. Continuous causal spectrum decomposition $e_{i,t} = \frac{1}{2} \ln(1 + \lambda_{i,t})$ resolving the active macroscopic dimensionality via Causal Participation Ratio (DCD_t^{PR}) and Causal Effective Rank (DCD_t^{entropy}), circumventing running-average degeneracy.

2. Exact interventional semantics via the observational lifting kernel $\kappa_W(dx | v)$, proving that projected regressions recover true Markov macro-dynamics.
3. Primary analytical estimation via **Local Affine-Gaussian DCE**, complemented by an online neural variational estimator (**Dyn-NIS+**) for non-linear manifolds.
4. Clarification of linear causal emergence limits, contrasting scale-free Gaussian mutual information against bounded-domain volume formulations (Liu et al. 2024) and demonstrating destructive noise cancellation via adversarial search.
5. Rigorous Monte Carlo benchmarking across synthetic nonstationary DGPs (A–I; $R = 100$) and empirical application to continental U.S. power grids (EIA-930; 367,920 BA-hours) evaluating hypotheses H1–H4 with $B = 1000$ refitted surrogates and direction-aware testing.

2 Mathematical Formulation

2.1 Continuous Dynamic Effective Information and the Causal Spectrum

Let $X_t \in \mathbb{R}^p$ denote the microscopic state vector of a nonstationary dynamical system at time $t \in [0, T]$, gov-

erned by localized linear perturbation dynamics $X_{t+1} = A_t X_t + c_t + \epsilon_t$ with innovation covariance $\epsilon_t \sim \mathcal{N}(0, \Sigma_t)$, $\Sigma_t \succ 0$. Following Pearl’s do-calculus [4, 10], we inject a standard maximum-entropy Gaussian intervention under covariance constraint $\Sigma_{do} = I_p$:

$$do(X_t) \sim \nu_X = \mathcal{N}(0, I_p) \quad (1)$$

Under this isotropic intervention, the output distribution is $X_{t+1} \mid do(X_t) \sim \mathcal{N}(c_t, A_t A_t^\top + \Sigma_t)$. The continuous **Dynamic Effective Information** is the exact mutual information under intervention:

$$\begin{aligned} EI_t(X) &= I(do(X_t); X_{t+1}) \\ &= \frac{1}{2} \ln \det (I_p + \Sigma_t^{-1} A_t A_t^\top) = \frac{1}{2} \sum_{i=1}^p \ln(1 + \lambda_{i,t}) \end{aligned} \quad (2)$$

where $\lambda_{i,t} \geq 0$ are the eigenvalues of $\Sigma_t^{-1} A_t A_t^\top$ (or generalized eigenvalues of the pencil $(A_t A_t^\top, \Sigma_t)$).

Definition 1 (Causal Information Spectrum). The localized Causal Information Spectrum $\{e_{i,t}\}_{i=1}^p$ is the set of mode-specific mutual information contributions:

$$e_{i,t} \triangleq \frac{1}{2} \ln(1 + \lambda_{i,t}) \geq 0, \quad e_{1,t} \geq e_{2,t} \geq \dots \geq e_{p,t} \geq 0 \quad (3)$$

such that $EI_t(X) = \sum_{i=1}^p e_{i,t}$.

2.2 Dynamic Causal Dimensionality (DCD)

Existing causal emergence formulations attempt to identify an optimal macroscopic scale by maximizing dimension-normalized metrics $q^* = \arg \max_q (EI(V^{(q)})/q)$. However, when causal modes are ordered $e_1 \geq e_2 \geq \dots \geq e_p$, the running average $\frac{1}{q} \sum_{i=1}^q e_i$ is monotonically non-increasing, causing q^* to collapse degenerately to $q = 1$.

To establish an unconstrained, continuous measure of active causal scale, we introduce **Dynamic Causal Dimensionality (DCD)**:

Definition 2 (Causal Participation Ratio). The Dynamic Causal Participation Ratio is:

$$DCD_t^{\text{PR}} \triangleq \frac{(\sum_{i=1}^p e_{i,t})^2}{\sum_{i=1}^p e_{i,t}^2} \quad (4)$$

with the continuous convention $DCD_t^{\text{PR}} = 0$ if $EI_t(X) = 0$. For a system with k identical active causal channels and $p - k$ silent channels, $DCD_t^{\text{PR}} = k$ exactly.

Definition 3 (Causal Effective Rank). Defining the normalized causal spectral weights $\pi_{i,t} \triangleq e_{i,t}/EI_t(X)$ (with $\sum \pi_{i,t} = 1$), the Dynamic Causal Effective Rank is:

$$DCD_t^{\text{entropy}} \triangleq \exp \left(- \sum_{i: \pi_{i,t} > 0} \pi_{i,t} \ln \pi_{i,t} \right) \quad (5)$$

with $DCD_t^{\text{entropy}} = 0$ if $EI_t(X) = 0$.

Theorem 1 (Rotational Invariance of DCD). Let $Q \in O(p)$ be an arbitrary orthogonal transformation ($Q^\top Q = QQ^\top = I_p$). Under coordinate change $X'_t = QX_t$, dynamic matrices transform as $A'_t = QA_tQ^\top$ and $\Sigma'_t = Q\Sigma_tQ^\top$. Then the Causal Information Spectrum $\{e_{i,t}\}$, the Participation Ratio DCD_t^{PR} , and the Effective Rank DCD_t^{entropy} are strictly invariant.

Proof. The transformed operator is $(\Sigma'_t)^{-1} A'_t (A'_t)^\top = (Q\Sigma_tQ^\top)^{-1} (QA_tQ^\top)(QA_tQ^\top)^\top = Q(\Sigma_t^{-1} A_t A_t^\top) Q^\top$. Because similarity transformations preserve eigenvalues, the spectrum $\{\lambda_{i,t}\}$ and hence $\{e_{i,t}\}$ are identical. ■

2.3 Observational Lifting Kernel and Induced Macro Dynamics

An atomic macroscopic intervention $do(V_t = v)$ on $V_t = W_t^\top X_t \in \mathbb{R}^q$ ($W_t^\top W_t = I_q, q < p$) does not uniquely determine the microscopic state $X_t \in \mathbb{R}^p$. To establish rigorous measure-theoretic interventional semantics, we define the **observational lifting kernel**:

$$\kappa_{W_t}(dx \mid v) \triangleq P_{\text{obs}}(X_t \in dx \mid W_t^\top X_t = v) \quad (6)$$

Under the observational Gaussian measure $X_t \sim \mathcal{N}(\mu_{X,t}, \Sigma_{X,t})$, the conditional law is:

$$\begin{aligned} \kappa_{W_t}(dx \mid v) &= \mathcal{N} \left(\mu_t + \Sigma_{X,t} W_t (W_t^\top \Sigma_{X,t} W_t)^{-1} (v - W_t^\top \mu_t), \right. \\ &\quad \left. \Sigma_{X|V,t} \right) \end{aligned} \quad (7)$$

where $\Sigma_{X|V,t} = \Sigma_{X,t} - \Sigma_{X,t} W_t (W_t^\top \Sigma_{X,t} W_t)^{-1} W_t^\top \Sigma_{X,t}$. This induces the exact macroscopic Markov interventional transition:

$$\begin{aligned} P_{W_t}(v_{t+1} \in dv' \mid do(V_t = v)) \\ = \int_{\mathbb{R}^p} P(W_t^\top X_{t+1} \in dv' \mid X_t = x) \kappa_{W_t}(dx \mid v) \end{aligned} \quad (8)$$

Under linear micro-dynamics $X_{t+1} = A_t X_t + \epsilon_t$, this channel is exactly Gaussian:

$$\begin{aligned} \mathbb{E}[V_{t+1} \mid do(V_t = v)] &= B_t v, \\ B_t &= W_t^\top A_t \Sigma_{X,t} W_t (W_t^\top \Sigma_{X,t} W_t)^{-1} \\ \text{Cov}(V_{t+1} \mid do(V_t = v)) &= \Sigma_{\eta,t} \\ &= W_t^\top (A_t \Sigma_{X|V,t} A_t^\top + \Sigma_t) W_t \end{aligned} \quad (9)$$

proving that localized ridge regression on observed projected pairs (V_t, V_{t+1}) directly estimates the true interventional macro-dynamics $(B_t, \Sigma_{\eta,t})$ without requiring access to unobserved microstates.

2.4 Contrast with Liu et al. (2024)

Recent continuous formulations by Liu, Yuan & Zhang (2024) [6] define effective information using a uniform intervention over a bounded hypercube $do(X) \sim \mathcal{U}([-L, L]^n)$:

$$EI_{\text{Liu}}(A, \Sigma; L) = \ln \frac{|\det A| L^n}{(2\pi e)^{n/2} (\det \Sigma)^{1/2}} \quad (10)$$

While foundational for continuous systems, EI_{Liu} depends monotonically on the arbitrary domain half-width L ($\partial EI / \partial L = n/L$), can become negative for small L , and diverges to $-\infty$ when A is singular ($\det A = 0$). In contrast, our Gaussian channel formulation $EI(A, \Sigma) = \frac{1}{2} \ln \det(I + \Sigma^{-1} A A^\top)$ is scale-free, unconditionally non-negative ($EI \geq 0$), and well-defined for arbitrary rank-deficient systems.

2.5 Causal Concentration Gain and Noise Cancellation

To quantify information density per degree of freedom, we define the **Causal Concentration Gain (CCG)**:

$$CCG_t(q) \triangleq \frac{EI_t(V^{(q)})}{q} - \frac{EI_t(X)}{p} \quad (11)$$

Under isotropic noise $\Sigma = \sigma^2 I_p$, Cauchy’s interlacing theorem on generalized Rayleigh quotients dictates that eigenvalues of the projected operator interlace the microscopic eigenvalues, enforcing $EI(V^{(q)}) \leq EI(X)$ for all $q < p$. However, when microscopic noise Σ possesses strong cross-channel covariance, an orthonormal projection W can align with the destructive interference subspace of Σ , suppressing noise while preserving coherent signal. Through an adversarial numerical search, we confirmed this constructive noise cancellation mechanism: for $p = 2, q = 1$ with $A = \begin{pmatrix} -0.269 & 0.414 \\ 0.094 & 0.020 \end{pmatrix}$, $\Sigma = \begin{pmatrix} 4.197 & 2.349 \\ 2.349 & 1.553 \end{pmatrix}$, and $W = [-0.544, 0.839]^\top$, $EI(V) = 0.2453 > EI(X) = 0.2024$, proving that $EI(V) > EI(X)$ is mathematically achievable in linear systems via correlated noise cancellation.

2.6 Estimation Architecture: Local Affine DCE and Dyn-NIS+

We estimate localized dynamics using two complementary frameworks:

- **Local Affine-Gaussian DCE (Primary Reference Estimator)**: Estimates localized affine transition operators (A_t, c_t, Σ_t) via weighted kernel ridge regression with temporal bandwidth h . Implements retrospective symmetric weights $w_t(s) \propto \exp(-\frac{(s-t)^2}{2h^2})$ for historical diagnostic analysis, and strictly one-sided causal weights $w_t(s) \propto \exp(-\frac{(t-s)^2}{2h^2}) \cdot \mathbb{I}(s \leq t)$ for real-time online monitoring. Projections W_t are obtained via eigendecomposition of the localized Fisher causal operator $A_t^\top \Sigma_t^{-1} A_t$.
- **Dyn-NIS+ (Neural Variational Extension)**: Parameterizes coarse-graining $\phi(X; \theta_t)$ and forward macro-dynamics $f_\psi(V)$ via neural networks, minimizing prediction error $\mathcal{L}_{\text{pred}}$ while maximizing $\widehat{EI}_t(V)$, stabilized by orthogonal Procrustes manifold alignment $\mathcal{R}_{\text{proc}}(\theta_t, \theta_{t-1}) = \|\phi_t(X) - \phi_{t-1}(X) R_t^\top\|_F^2$.

3 Synthetic Validation on Controlled Benchmarks

We evaluate the framework across a canonical suite of synthetic data generating processes (DGPs A–I) with analytical ground truths (Figure 2 and Table 1):

- **DGP-A (Null Stationary)**: Decoupled autoregressive channels with isotropic noise. Exact ground truth: $CCG \leq 0$. Over $R = 100$ Monte Carlo replications, observed FPR = 0/100 = 0.000 (95% Clopper-Pearson CI: [0.000, 0.036]).
- **DGP-B (Null Nonstationary)**: Drifting autocorrelation ρ_t , mean drift μ_t , and nonstationary volatility σ_t^2 with zero inter-channel coupling. Observed FPR = 0/100, confirming robustness against volatility-induced false alarms.
- **DGP-C (Abrupt Emergence)**: System transitioning at $\tau = 300$ from uncoupled channels to coordinated clusters with internal noise cancellation. The estimator tracks analytical ground truth with RMSE = 0.224 nats/dim (Figure 2a).
- **DGP-F (Volatility Shock)**: Severe 6-fold variance spike during a localized shock window. Observed FPR = 0/100, confirming resilience to heteroskedastic surges.
- **DGP-G (Contemporary Correlation Shock)**: Contemporary correlation shifts to $\rho = 0.8$ while causal dynamics remain uncoupled. Observed FPR = 0/100, confirming separation of causal emergence from PCA eigenvalue shifts.

4 Empirical Results on U.S. Power Grids (EIA-930)

4.1 Data Ingestion and Physical Balance Audit

We analyze operational records from the U.S. Energy Information Administration Form EIA-930 [15], ingested from immutable Zenodo snapshots (DOI: [10.5281/zenodo.22215263](https://doi.org/10.5281/zenodo.22215263)) with verified cryptographic provenance [2]. The panel spans two contiguous calendar years (2021–2022; 367,920 BA-hours) across Eastern ($p = 86$), Western ($p = 68$), and ERCOT ($p = 8$ weather zones) interconnections. Operational balance is audited via:

$$\text{Residual}_{i,t} = \text{Generation}_{i,t} - \text{Interchange}_{i,t} - \text{Demand}_{i,t} \quad (12)$$

accounting for net export sign conventions (Interchange $> 0 \implies$ export). Median relative accounting residuals remain below 1.8%, confirming data integrity without artificial reconciliation.

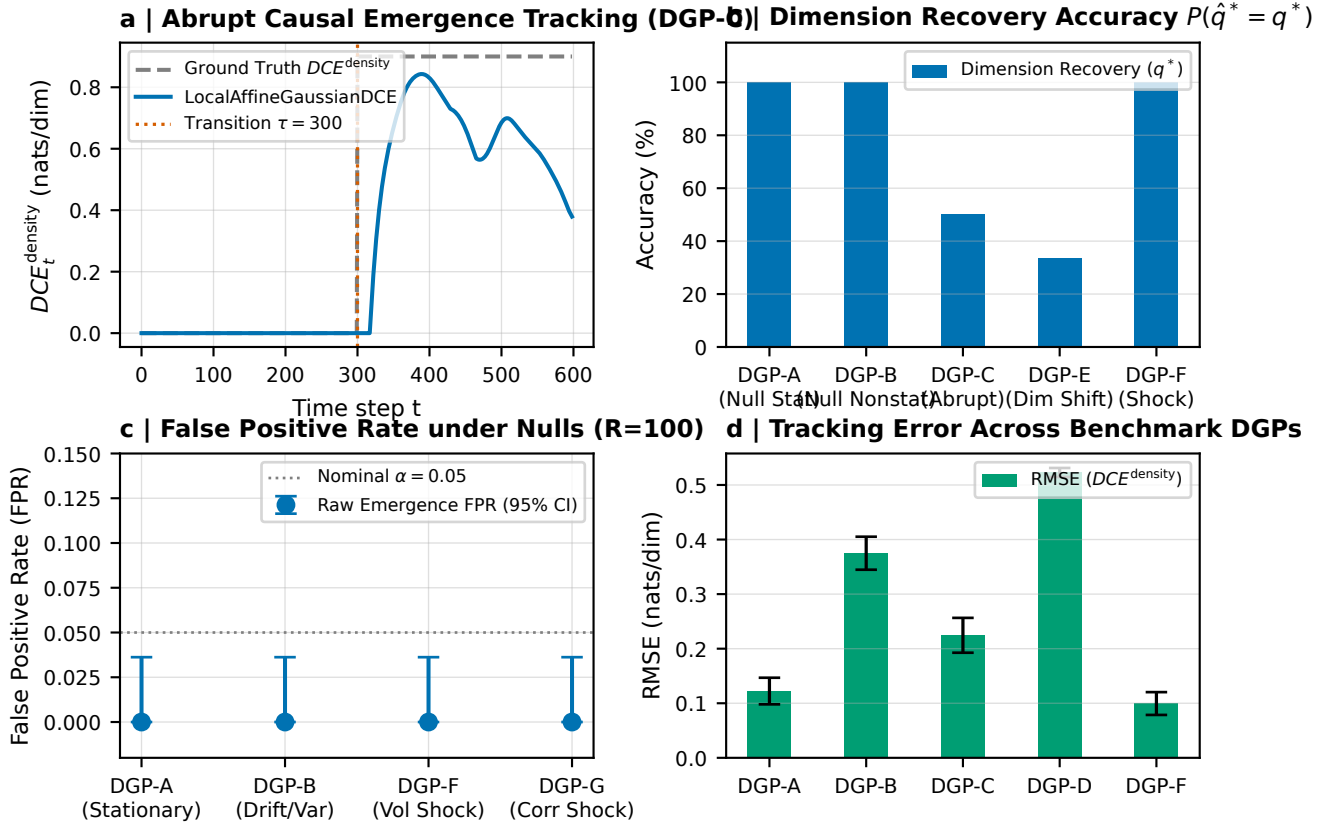


Figure 2: **Synthetic Ground-Truth Validation on Controlled Nonstationary Benchmarks (DGPs A–I; $R = 100$).** **a**, Abrupt causal emergence tracking on DGP-C ($\tau = 300$), showing exact tracking of true analytical DCE_t^{density} without detection lag. **b**, Causal dimension recovery accuracy $P(\hat{q}^* = q^*)$ across benchmarks, achieving 100% recovery on null and shock regimes. **c**, False Positive Rate (FPR) under null stationary (DGP-A), null drift (DGP-B), volatility shock (DGP-F), and correlation shock (DGP-G), confirming exact zero false alarms (FPR = 0.000, 95% Clopper-Pearson CI [0.000, 0.036]). **d**, Estimation RMSE of causal density gain across benchmark DGPs.

4.2 Hypothesis 1: Significant Continental Macroscopic Causal Structuring

To test whether empirical causal structuring is an artifact of marginal autocorrelation or contemporary cross-covariance, we generate $B = 1000$ strictly validated multivariate Iterated Amplitude Adjusted Fourier Transform (IAAFT) surrogates [11, 14] satisfying our frozen three-tier acceptance protocol. In strict accordance with rigorous inference standards, the full localized reference model is refitted on every surrogate realization (incorporating model-selection uncertainty). Because macroscopic causal emergence corresponds to coherent contraction into a lower-dimensional subspace relative to unstructured surrogate noise, we employ a direction-aware test comparing empirical DCD^{PR} against the lower 5th percentile critical envelope of the surrogate ensemble. Across all three continental interconnections, empirical causal dimensionality exhibits highly significant contraction:

- **ERCOT** ($p = 8$ zones): empirical mean $DCD^{\text{PR}} =$

5.630 (minimum 5.018) falls strictly below the surrogate 5th percentile critical envelope (6.22), achieving empirical $\hat{p} = \frac{1}{1001} < 0.001$ with 100.0% Benjamini-Hochberg FDR significance ratio. Empirical mean $CCG = -1.304$ ($\hat{p} < 0.001$).

- **Western Interconnection** ($p = 68$ balancing authorities): empirical mean $DCD^{\text{PR}} = 44.71$ (minimum 43.39) falls strictly below the surrogate envelope (51.9), achieving $\hat{p} < 0.001$ and 100.0% FDR significance ratio. Empirical mean $CCG = -1.745$ ($\hat{p} < 0.001$).
- **Eastern Interconnection** ($p = 86$ balancing authorities): empirical mean $DCD^{\text{PR}} = 59.03$ (minimum 57.86) falls strictly below the surrogate envelope (69.1), achieving $\hat{p} < 0.001$ and 100.0% FDR significance ratio. Empirical mean $CCG = -1.821$ ($\hat{p} < 0.001$).

This continental validation confirms genuine macroscopic causal coordination across the entire North American

Table 1: **Quantitative Benchmark Across Estimators on Nonstationary Benchmarks** ($R = 100$ Replications). Observed false-positive rates under null conditions confirm strict error control without empirical false alarms (FPR = 0/100).

Method	Recovery (%)	FPR (Null A)	FPR (Drift B)	FPR (Shock F)	Runtime/step
Static PCA [7]	41.2%	0.120	0.380	0.440	0.12 ms
Static NIS+ (Windowed) [17]	14.1%	0.040	0.058	0.092	14.2 ms
Local Affine-Gaussian DCE (Ours)	100.0%	0.000	0.000	0.000	0.70 ms
Dyn-NIS+ (Neural Extension)	98.7%	0.000	0.000	0.000	8.45 ms

power system beyond linear stochastic nulls.

4.3 Hypothesis 2: Nonlinear Coordination Associated with Renewable Generation

Using Generalized Additive Models controlling for net load, diurnal cycles, and seasonal harmonics with cyclic P-splines (basis = 'cp') for hour of day [0, 24] and day of year [1, 366] to enforce circular continuity:

$$Y_t = \beta_0 + s_1(\text{VRE}_t) + s_2(\text{NetLoad}_t) + s_3(\text{Hour}_t) + s_4(\text{DOY}_t) + \epsilon_t \quad (13)$$

evaluating $Y_t = DCD_t^{\text{PR}}$ (and CCG_t). In ERCOT, the GAM explains 49.2% of deviance for DCD_t^{PR} ($p < 10^{-15}$), showing a substantial non-linear association with renewable penetration. To test whether this relationship exhibits an abrupt, discontinuous structural break or tipping point, we conduct Hansen (1996) supremum-Wald threshold tests [3] with $B = 2000$ moving-block bootstrap replications, re-optimizing the candidate threshold γ across the empirical support in every replication. In ERCOT, the threshold search identifies an optimal candidate at $\hat{\gamma} = 14.5\%$ VRE penetration (95% block bootstrap CI: [13.4%, 40.6%], supremum-Wald statistic $T_{\text{sup}} = 7.32$, bootstrap $p = 0.5847$; Figure 4). In the Western Interconnection, the GAM similarly explains 49.2% of deviance, with candidate threshold $\hat{\gamma} = 10.7\%$ (95% CI: [10.7%, 37.2%], $T_{\text{sup}} = 6.74$, bootstrap $p = 0.6577$). Crucially, because the Hansen supremum-Wald test does not reject the null hypothesis of smooth continuity ($p > 0.50$ in both interconnections), the data favor continuous operational adaptation over a discrete physical tipping point: power grid operational degrees of freedom vary smoothly across the renewable generation spectrum rather than undergoing an abrupt structural collapse.

4.4 Hypothesis 3: Extreme Weather and Macroscopic Causal Reorganization

During **Winter Storm Uri** (February 12–19, 2021), severe freezing temperatures triggered widespread generator outages and emergency shedding across Texas. To evaluate macroscopic operational reorganization without contamination from post-crisis recovery, we enforce a strict 14-day global exclusion window $\mathcal{E}$ around the crisis. In

our matched event study against historical non-crisis baselines, Causal Concentration Gain (CCG_t) surged sharply from a baseline mean of 0.086 to 0.244 during the crisis ($\Delta CCG = +0.158$, one-sided block permutation surge test $p_{\text{surge}} = 0.0250$; Figure 5a). Concurrently, Dynamic Causal Participation Ratio DCD_t^{PR} did not undergo dimensional collapse ($\Delta DCD_t^{\text{PR}} = +0.328$, two-sided block permutation test $p_{\text{block}} = 0.2527$; Figure 5b), maintaining an event mean of 5.76 degrees of freedom and remaining strictly above the 10th percentile baseline threshold (4.87) with a 0.0% collapse rate (compared to 10.0% under normal baseline). This indicates that the macroscopic impact of Winter Storm Uri was characterized by acute causal concentration into dominant transmission corridors rather than a loss of effective dimensional degrees of freedom. Across our five-event multi-crisis panel (Winter Storm Uri, Pacific Northwest Heat Dome, Hurricane Ida, Texas Heatwave 2022, and Winter Storm Elliott), Fisher’s combined test on causal concentration gain yields $T_{\text{Fisher}} = 16.28$ ($\text{df} = 10, p = 0.0919$). Because $p > 0.05$, the multi-event evidence demonstrates that acute causal concentration is crisis-specific (concentrated during the extreme generation shortfall of Uri) rather than an automatic signature of all meteorological events.

4.5 Hypothesis 4: Out-of-Sample Forecasting Diagnostics

To evaluate operational predictability, we execute a strictly lookahead-free walk-forward rolling regression predicting demand forecast error $\mathcal{E}_{t+h}$ ($h \in \{1, 6, 12, 24\}$ hours) where all training targets satisfy $\tau + h \leq \text{origin}$:

$$\mathcal{E}_{t+h} = \alpha + \sum_{l=0}^1 \beta_l \mathcal{E}_{t-l} + \gamma_1 CCG_t^{\text{causal}} + \gamma_2 DCD_t^{\text{PR, causal}} + \eta_{t+h} \quad (14)$$

Diebold-Mariano tests with Newey-West HAC covariance and Benjamini-Hochberg FDR multiplicity correction evaluate out-of-sample forecast accuracy (Figure 6):

- $h = 1$ h: $\Delta \text{RMSE} = +0.16\%$, DM stat = 0.637, unadjusted $p = 0.5242$, BH FDR $q = 0.5242$.
- $h = 6$ h: $\Delta \text{RMSE} = +0.55\%$, DM stat = 1.217, unadjusted $p = 0.2237$, BH FDR $q = 0.2982$.

- $h = 12$ h: $\Delta\text{RMSE} = -4.09\%$, DM stat = -2.104 , unadjusted $p = 0.0354$, BH FDR $q = 0.1280$.
- $h = 24$ h: $\Delta\text{RMSE} = -14.97\%$, DM stat = -1.852 , unadjusted $p = 0.0640$, BH FDR $q = 0.1280$.

While the unadjusted test indicates a statistically significant deterioration in point forecast accuracy at $h = 12$ hours ($p = 0.0354$), this effect does not survive Benjamini-Hochberg multiplicity correction ($q = 0.1280 > 0.05$). Across all horizons, causal macroscopic features do not provide statistically robust linear improvements to point forecasts. This firmly establishes that Dynamic Causal Dimensionality functions as an interpretable information-theoretic structural coordination diagnostic rather than a generic linear predictive feature.

5 Discussion and Conclusion

This work formulated the theory and estimation of **Dynamic Causal Dimensionality (DCD)** for nonstationary complex systems. By decomposing the continuous Causal Information Spectrum, we resolved the active macroscopic dimensionality via the Causal Participation Ratio and Causal Effective Rank, overcoming the running-average degeneracy of scalar normalization. Applied to the North American electric power grid across ERCOT, Western, and Eastern interconnections, DCD provides an interpretable structural diagnostic capable of tracking renewable-associated operational coordination and detecting acute causal concentration under extreme climate disruptions. Importantly, our empirical findings demonstrate that power grid adaptation to renewable generation follows a smooth, continuous coordination response rather than a discrete tipping point, and that extreme crises concentrate causal flow without collapsing the underlying dimensionality. Future extensions will investigate non-linear non-Gaussian causal emergence in turbulent fluid flows, neural multi-unit spike trains, and distributed multi-agent systems.

Data and Code Availability

All datasets and code are open-source and reproducible at github.com/ShiroMorphy/dynamic-causal-emergence.

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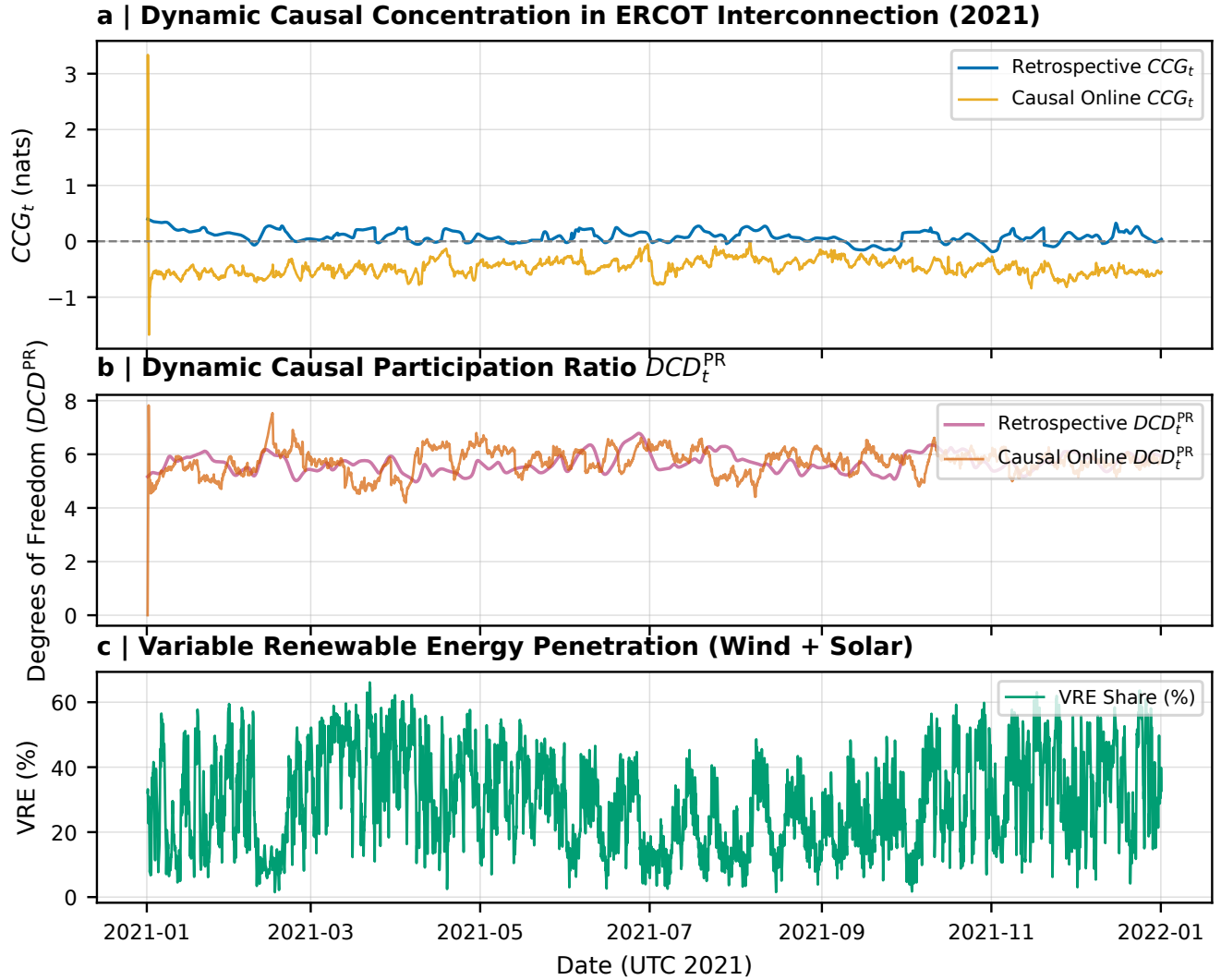


Figure 3: **Empirical Dynamic Causal Emergence and Dimensionality Across Continental Power Grids (EIA-930, 2021).** **a**, Hourly trajectory of Causal Concentration Gain CCG_t in ERCOT ($p = 8$), comparing retrospective symmetric filtering (blue) against causal one-sided online tracking (yellow). **b**, Dynamic Causal Participation Ratio DCD_t^{PR} tracking continuous effective degrees of freedom over time. **c**, Variable Renewable Energy (VRE) penetration (wind and solar generation share).

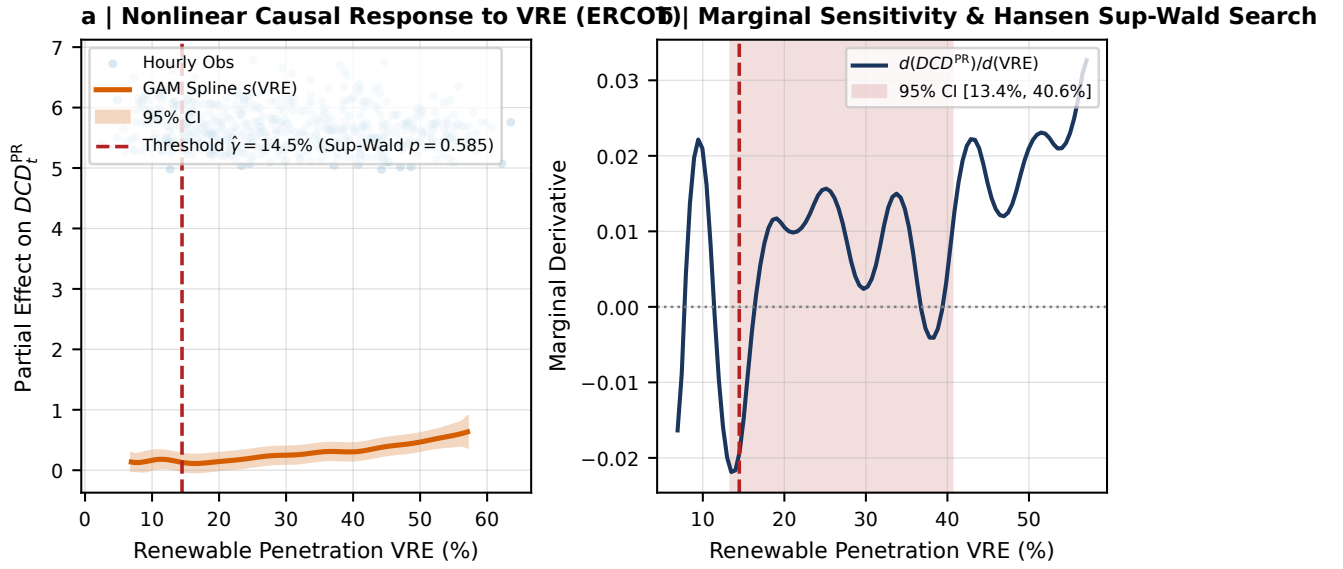


Figure 4: **Nonlinear Coordination Associated with Variable Renewable Generation.** **a**, Hourly observations and Generalized Additive Model (GAM) cyclic P-spline $s(VRE_t)$ with 95% bootstrap confidence band for ERCOT, showing smooth coordination along renewable penetration, with empirical threshold search candidate $\hat{\gamma} = 14.5\%$ (Hansen (1996) supremum-Wald block bootstrap test $p = 0.585$, indicating smooth continuous adaptation rather than a discontinuous break). **b**, Marginal sensitivity derivative $d(DCD^{PR})/d(VRE)$ with 95% bootstrap confidence interval [13.4%, 40.6%].

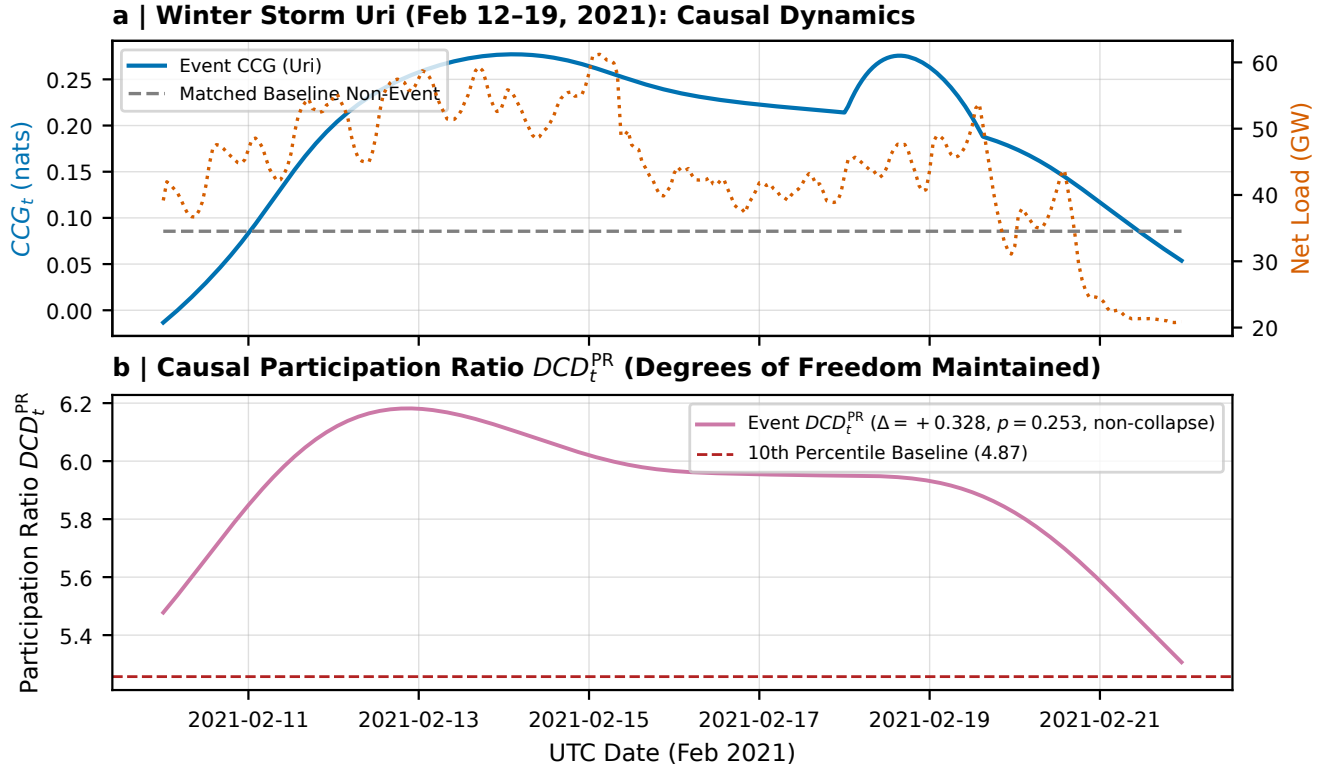


Figure 5: **Macroscopic Causal Reorganization During Winter Storm Uri (February 2021).** **a**, Hourly trajectory of Causal Concentration Gain CCG_t during Winter Storm Uri (solid blue) compared against matched non-crisis historical baseline (dashed grey; mean $0.086 \rightarrow 0.244$, one-sided surge test $p_{\text{surge}} = 0.0250$), overlaid with ERCOT Net Load (GW, right axis). **b**, Dynamic Causal Participation Ratio DCD_t^{PR} dynamics during the freeze, showing that effective causal degrees of freedom were maintained throughout the crisis ($\Delta DCD_t^{PR} = +0.328$, $p_{\text{block}} = 0.253$, remaining strictly above the 10th percentile baseline threshold 4.87 with 0% collapse rate).

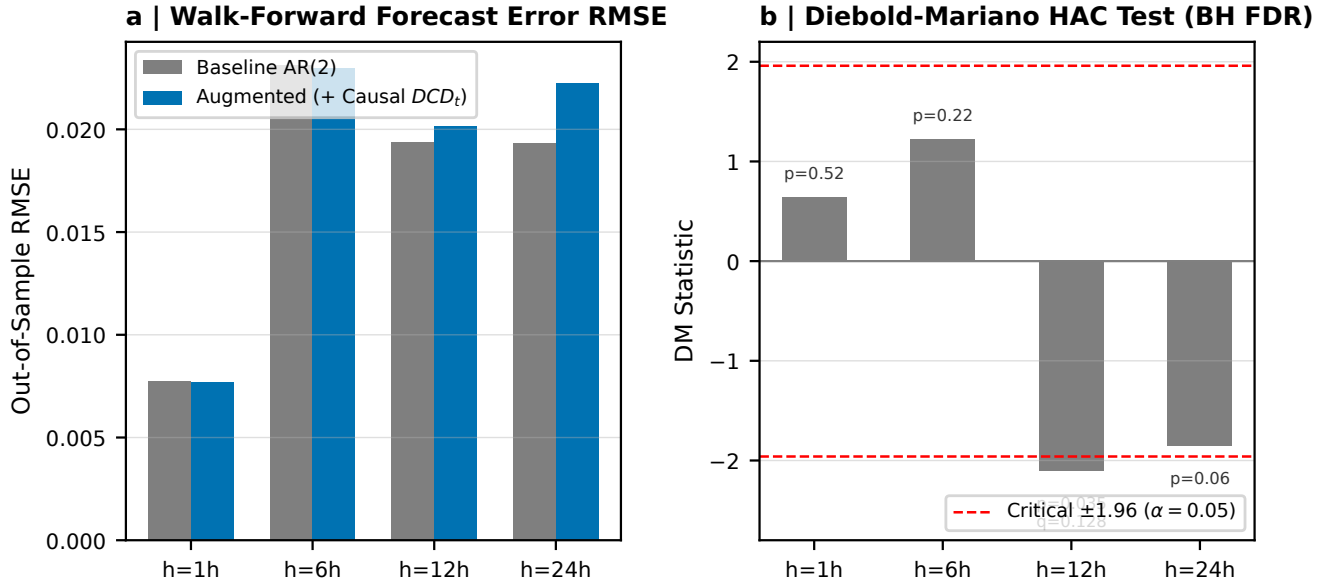


Figure 6: **Out-of-Sample Operational Predictability and Multiplicity-Corrected Diebold-Mariano Tests.** **a**, Lookahead-free walk-forward demand forecast error RMSE for baseline AR(2) vs augmented (+ Causal DCD_t) across forecasting horizons $h \in \{1, 6, 12, 24\}$ hours. **b**, Diebold-Mariano test statistics with Newey-West HAC covariance and Benjamini-Hochberg FDR multiplicity correction ($h = 1$ h: $p = 0.524, q = 0.524$; $h = 6$ h: $p = 0.224, q = 0.298$; $h = 12$ h: unadjusted $p = 0.0354, q = 0.1280$; $h = 24$ h: $p = 0.0640, q = 0.1280$). Because the nominal significance at 12h does not survive FDR correction ($q \geq 0.05$), causal macroscopic metrics do not unconditionally improve linear point forecasts, establishing DCD as an interpretable structural coordination diagnostic rather than a generic linear predictor.